CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Worked solutions 13 questions · 115 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

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1. **(a)** By expanding $(\cos\theta + i\sin\theta)^3$ and using de Moivre's theorem, show that

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$ **[4]**

(b) Hence find all values of θ, where $0 \leq \theta \leq \pi$, that satisfy

$$4\cos^3\theta - 3\cos\theta = \cos 2\theta$$

giving your answers as exact multiples of π. **[4]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$$

Expand the left-hand side by the binomial theorem. Only the real terms are needed. These are the terms with even powers of $i\sin\theta$:

$$\cos^3\theta + 3\cos\theta\,(i\sin\theta)^2 = \cos^3\theta - 3\cos\theta\sin^2\theta$$

Equating real parts and using $\sin^2\theta = 1 - \cos^2\theta$:

$$\begin{aligned}\cos 3\theta &= \cos^3\theta - 3\cos\theta\big(1 - \cos^2\theta\big)\\ &= \cos^3\theta - 3\cos\theta + 3\cos^3\theta\\ &= 4\cos^3\theta - 3\cos\theta\end{aligned}$$

as required.

(b) Using part (a), the equation becomes

$$\cos 3\theta = \cos 2\theta$$

Equal cosines give, for $n \in \mathbb{Z}$,

$$3\theta = 2n\pi \pm 2\theta$$

Taking the $+$ sign:

$$\theta = 2n\pi$$

In $0 \leq \theta \leq \pi$, this gives only $\theta = 0$.

Taking the $-$ sign:

$$5\theta = 2n\pi \quad\Longrightarrow\quad \theta = \frac{2n\pi}{5}$$

In $0 \leq \theta \leq \pi$, the possible integers are $n = 0, 1, 2$.

So the solutions are

$$\theta = 0, \qquad \theta = \frac{2\pi}{5}, \qquad \theta = \frac{4\pi}{5}$$

2. Use de Moivre's theorem, by expanding $(\cos\theta + i\sin\theta)^4$ and equating real parts, to express $\cos 4\theta$ in the form $a\cos^4\theta + b\cos^2\theta + c$, where a, b and c are integers to be found. **[5]**

Solution

By de Moivre's theorem:

$$\cos 4\theta + i\sin 4\theta = (\cos\theta + i\sin\theta)^4$$

In the binomial expansion of the right-hand side, the real terms are those containing even powers of $i\sin\theta$:

$$\begin{aligned}\mathrm{Re}\left[(\cos\theta + i\sin\theta)^4\right] &= \cos^4\theta + \binom{4}{2}\cos^2\theta\,(i\sin\theta)^2 + (i\sin\theta)^4 \\ &= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta\end{aligned}$$

Equating real parts and using $\sin^2\theta = 1 - \cos^2\theta$:

$$\begin{aligned}\cos 4\theta &= \cos^4\theta - 6\cos^2\theta\left(1 - \cos^2\theta\right) + \left(1 - \cos^2\theta\right)^2 \\ &= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta \\ &= 8\cos^4\theta - 8\cos^2\theta + 1\end{aligned}$$

This is in the form $a\cos^4\theta + b\cos^2\theta + c$ with $a = 8$, $b = -8$, $c = 1$.

3. Let $z = \cos\theta + i\sin\theta$.
You may assume that, for any positive integer n,

$$z^n - \frac{1}{z^n} = 2i\sin n\theta$$

By considering $\left(z - \frac{1}{z}\right)^3$, find constants A and B such that

$$\sin^3\theta = A\sin\theta + B\sin 3\theta$$ **[4]**

Solution

With $n = 1$, $z - \frac{1}{z} = 2i\sin\theta$, so

$$\left(z - \frac{1}{z}\right)^3 = (2i\sin\theta)^3 = -8i\sin^3\theta$$

Expanding with the binomial theorem and grouping the terms in pairs:

$$\begin{aligned}\left(z - \frac{1}{z}\right)^3 &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} \\ &= \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right) \\ &= 2i\sin 3\theta - 6i\sin\theta\end{aligned}$$

Equating the two expressions gives

$$-8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta \quad\Longrightarrow\quad \sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$$

so $A = \frac{3}{4}$ and $B = -\frac{1}{4}$.

4. **(a)** Use de Moivre's theorem to find the constants A, B and C such that

$$\cos^5\theta \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos^5\theta \,\mathrm{d}\theta \qquad \textbf{[3]}$$

Solution

(a) Let $z = \cos\theta + i\sin\theta$. By de Moivre's theorem, $z^n = \cos n\theta + i\sin n\theta$ and $z^{-n} = \cos n\theta - i\sin n\theta$, so

$$z^n + z^{-n} = 2\cos n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$. Expanding with the binomial theorem:

$$\begin{aligned}(2\cos\theta)^5 &= \left(z + z^{-1}\right)^5 \\ 32\cos^5\theta &= z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5} \\ &= \left(z^5 + z^{-5}\right) + 5\left(z^3 + z^{-3}\right) + 10\left(z + z^{-1}\right) \\ &= 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta\end{aligned}$$

Dividing by 32:

$$\cos^5\theta \equiv \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta$$

so $A = \frac{1}{16}$, $B = \frac{5}{16}$ and $C = \frac{5}{8}$.

(b) Using part (a):

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \cos^5\theta \,\mathrm{d}\theta &= \int_0^{\frac{\pi}{3}} \left(\frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta\right)\mathrm{d}\theta \\ &= \left[\frac{1}{80}\sin 5\theta + \frac{5}{48}\sin 3\theta + \frac{5}{8}\sin\theta\right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{80}\sin\frac{5\pi}{3} + \frac{5}{48}\sin\pi + \frac{5}{8}\sin\frac{\pi}{3} - 0 \\ &= \frac{1}{80}\left(-\frac{\sqrt{3}}{2}\right) + 0 + \frac{5}{8}\left(\frac{\sqrt{3}}{2}\right) \\ &= -\frac{\sqrt{3}}{160} + \frac{50\sqrt{3}}{160} \\ &= \frac{49\sqrt{3}}{160}\end{aligned}$$

5. **(a)** By expanding $(\cos\theta + i\sin\theta)^5$, use de Moivre's theorem to show that

$$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$$ **[5]**

(b) Hence solve, for $0 \leq \theta \leq \pi$, the equation

$$16\sin^5\theta - 20\sin^3\theta + 5\sin\theta = \sin 2\theta$$

giving your answers as exact multiples of π. **[5]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

In the binomial expansion of the left-hand side, the imaginary terms are those with odd powers of $i\sin\theta$:

$$\binom{5}{1}\cos^4\theta\,(i\sin\theta) + \binom{5}{3}\cos^2\theta\,(i\sin\theta)^3 + (i\sin\theta)^5$$

$$= 5i\cos^4\theta\sin\theta - 10i\cos^2\theta\sin^3\theta + i\sin^5\theta$$

Equating imaginary parts gives

$$\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$$

Substituting $\cos^2\theta = 1 - \sin^2\theta$:

$$\sin 5\theta = 5\sin\theta\big(1-\sin^2\theta\big)^2 - 10\sin^3\theta\big(1-\sin^2\theta\big) + \sin^5\theta$$

$$= 5\sin\theta - 10\sin^3\theta + 5\sin^5\theta - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta$$

$$= 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$$

as required.

(b) Using part (a), the equation becomes

$$\sin 5\theta = \sin 2\theta$$

Since $\sin A = \sin B$ when $A = B + 2k\pi$ or $A = \pi - B + 2k\pi$ for integer k, either

$$5\theta = 2\theta + 2k\pi \quad\Longrightarrow\quad \theta = \frac{2k\pi}{3}$$

which gives $\theta = 0$ and $\theta = \frac{2\pi}{3}$ in the range $(k = 0, 1)$, or

$$5\theta = \pi - 2\theta + 2k\pi \quad\Longrightarrow\quad \theta = \frac{(2k+1)\pi}{7}$$

which gives $\theta = \frac{\pi}{7}, \frac{3\pi}{7}, \frac{5\pi}{7}, \pi$ in the range $(k = 0, 1, 2, 3)$.

The solutions for $0 \leq \theta \leq \pi$ are therefore

$$\theta = 0,\ \frac{\pi}{7},\ \frac{3\pi}{7},\ \frac{2\pi}{3},\ \frac{5\pi}{7},\ \pi$$

6. **(a)** Use de Moivre's theorem to show that

$$\cos^6\theta = a\cos 6\theta + b\cos 4\theta + c\cos 2\theta + d$$

where a, b, c and d are constants to be found. **[5]**

(b) Hence show that

$$\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta = \frac{11}{48} + \frac{5\pi}{64}$$ **[4]**

Solution

(a) Let $z = \cos\theta + i\sin\theta$. By de Moivre's theorem:

$$z^n + z^{-n} = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$, so

$$(z + z^{-1})^6 = 64\cos^6\theta$$

Expanding by the binomial theorem:

$$(z + z^{-1})^6 = z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}$$

Group the terms in pairs:

$$\begin{aligned} 64\cos^6\theta &= (z^6 + z^{-6}) + 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) + 20 \\ &= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20 \end{aligned}$$

Dividing by 64:

$$\begin{aligned} \cos^6\theta &= \frac{1}{32}\cos 6\theta + \frac{12}{64}\cos 4\theta + \frac{30}{64}\cos 2\theta + \frac{20}{64} \\ &= \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16} \end{aligned}$$

Hence

$$a = \frac{1}{32}, \qquad b = \frac{3}{16}, \qquad c = \frac{15}{32}, \qquad d = \frac{5}{16}$$

(b) Using the identity from part (a):

$$\cos^6\theta = \frac{1}{32}\big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)$$

So

$$\begin{aligned} \int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\int_0^{\pi/4} \big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{6}{4}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \end{aligned}$$

At $\theta = \frac{\pi}{4}$:

$$\sin\frac{3\pi}{2} = -1, \qquad \sin\pi = 0, \qquad \sin\frac{\pi}{2} = 1$$

Therefore

$$\begin{aligned}
\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\left(-\frac{1}{6} + \frac{3}{2}\times 0 + \frac{15}{2}\times 1 + 10\times\frac{\pi}{4}\right) \\
&= \frac{1}{32}\left(-\frac{1}{6} + \frac{15}{2} + \frac{5\pi}{2}\right) \\
&= \frac{1}{32}\left(\frac{44}{6} + \frac{5\pi}{2}\right) \\
&= \frac{11}{48} + \frac{5\pi}{64}
\end{aligned}$$

7. A complex number z has modulus 1 and argument θ.

(a) Show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta, \qquad n \in \mathbb{Z}^+$$ **[3]**

(b) Hence, by expanding $\left(z + \frac{1}{z}\right)^3$ and $\left(z - \frac{1}{z}\right)^3$, show that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$ **[7]**

Solution

(a) Since $|z| = 1$ and $\arg z = \theta$, we have

$$z = \cos\theta + i\sin\theta$$

So, by de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta$$

and

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i\sin(-n\theta) = \cos n\theta - i\sin n\theta$$

Now add:

$$z^n + \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) + (\cos n\theta - i\sin n\theta) = 2\cos n\theta$$

Now subtract:

$$z^n - \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta) = 2i\sin n\theta$$

Therefore

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta$$

(b) From part (a), with $n = 1$:

$$z + \frac{1}{z} = 2\cos\theta$$

Cubing both sides gives

$$\left(z + \frac{1}{z}\right)^3 = (2\cos\theta)^3 = 8\cos^3\theta$$

Expand the left-hand side:

$$\begin{aligned}\left(z + \frac{1}{z}\right)^3 &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \\ &= \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)\end{aligned}$$

Using part (a) again:

$$8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$$

So

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

Now use part (a) with

$$z - \frac{1}{z} = 2i\sin\theta$$

Cubing both sides:

$$\left(z - \frac{1}{z}\right)^3 = (2i\sin\theta)^3 = -8i\sin^3\theta$$

Expand:

$$\begin{aligned}\left(z - \frac{1}{z}\right)^3 &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} \\ &= \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)\end{aligned}$$

Using part (a):

$$-8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta$$

Hence

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

Let $t = \tan\theta$. Then

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{\dfrac{\sin 3\theta}{\cos^3\theta}}{\dfrac{\cos 3\theta}{\cos^3\theta}} \\ &= \frac{\dfrac{3\sin\theta - 4\sin^3\theta}{\cos^3\theta}}{\dfrac{4\cos^3\theta - 3\cos\theta}{\cos^3\theta}} \\ &= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta} \\ &= \frac{3t(1+t^2) - 4t^3}{4 - 3(1+t^2)} \\ &= \frac{3t - t^3}{1 - 3t^2}\end{aligned}$$

Therefore

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

8. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2\cos q\pi$, where q is a rational number. **[4]**

Solution

(a) By de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

Expand the left-hand side using the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 &= \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta \\ &\quad + 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Now use

$$\sin^2\theta = 1 - \cos^2\theta \qquad \text{and} \qquad \sin^4\theta = (1 - \cos^2\theta)^2$$

So

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)^2 \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

(b) Let

$$x = 2\cos\theta$$

Then, using part (a):

$$\begin{aligned}x^5 - 5x^3 + 5x &= (2\cos\theta)^5 - 5(2\cos\theta)^3 + 5(2\cos\theta) \\ &= 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta \\ &= 2\cos 5\theta\end{aligned}$$

So the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

becomes

$$2\cos 5\theta + 1 = 0$$

that is

$$\cos 5\theta = -\frac{1}{2}$$

To obtain distinct values of $x = 2\cos\theta$, take $0 \leq \theta \leq \pi$. Then $0 \leq 5\theta \leq 5\pi$, and in this interval

$$\cos 5\theta = -\frac{1}{2}$$

when

$$5\theta = \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{8\pi}{3},\ \frac{10\pi}{3},\ \frac{14\pi}{3}$$

Hence

$$\theta = \frac{2\pi}{15},\ \frac{4\pi}{15},\ \frac{8\pi}{15},\ \frac{2\pi}{3},\ \frac{14\pi}{15}$$

Therefore the roots are

$$x = 2\cos\frac{2\pi}{15},\quad 2\cos\frac{4\pi}{15},\quad 2\cos\frac{8\pi}{15},\quad 2\cos\frac{2\pi}{3},\quad 2\cos\frac{14\pi}{15}$$

These are five distinct roots, so they are all the roots of the equation.

9. **(a)** By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big) \qquad \textbf{[5]}$$

(b) Hence, using the substitution $x = \tan\theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x \qquad \textbf{[6]}$$

Solution

(a) By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z - z^{-1} = 2i\sin\theta$, so

$$(z - z^{-1})^6 = (2i\sin\theta)^6 = 64i^6\sin^6\theta = -64\sin^6\theta$$

Now expand $(z - z^{-1})^6$ by the binomial theorem:

$$\begin{aligned}(z - z^{-1})^6 &= z^6 - 6z^5(z^{-1}) + 15z^4(z^{-1})^2 - 20z^3(z^{-1})^3 \\ &\quad + 15z^2(z^{-1})^4 - 6z(z^{-1})^5 + (z^{-1})^6 \\ &= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}\end{aligned}$$

Therefore

$$\begin{aligned}(z - z^{-1})^6 &= (z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20 \\ &= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20\end{aligned}$$

Equating this with the earlier result gives

$$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$$

so

$$64\sin^6\theta = 20 - 30\cos 2\theta + 12\cos 4\theta - 2\cos 6\theta$$

and hence

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

as required.

(b) Let

$$I = \int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x$$

Use the substitution

$$x = \tan\theta$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta \quad \Longrightarrow \quad \mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$$

and

$$1 + x^2 = 1 + \tan^2\theta = \sec^2\theta$$

Also, when $x = 0$, $\theta = 0$, and when $x = 1$, $\theta = \frac{\pi}{4}$.

So

$$\begin{aligned} I &= \int_0^{\pi/4} \frac{\tan^6\theta}{(\sec^2\theta)^4}\sec^2\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \frac{\tan^6\theta}{\sec^6\theta}\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \frac{\sin^6\theta}{\cos^6\theta}\cos^6\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \sin^6\theta\,\mathrm{d}\theta \end{aligned}$$

Using the result from part (a):

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

Therefore

$$\begin{aligned} I &= \frac{1}{32}\int_0^{\pi/4}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[10\theta - \frac{15}{2}\sin 2\theta + \frac{3}{2}\sin 4\theta - \frac{1}{6}\sin 6\theta\right]_0^{\pi/4} \end{aligned}$$

Now substitute the limits:

$$\sin\frac{\pi}{2} = 1, \qquad \sin\pi = 0, \qquad \sin\frac{3\pi}{2} = -1$$

so

$$\begin{aligned} I &= \frac{1}{32}\left(10\times\frac{\pi}{4} - \frac{15}{2}(1) + \frac{3}{2}(0) - \frac{1}{6}(-1)\right) \\ &= \frac{1}{32}\left(\frac{5\pi}{2} - \frac{15}{2} + \frac{1}{6}\right) \\ &= \frac{1}{32}\left(\frac{15\pi - 44}{6}\right) \\ &= \frac{15\pi - 44}{192} \end{aligned}$$

Hence the exact value is

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x = \frac{15\pi - 44}{192}$$

10. **(a)** Use de Moivre's theorem to show that $\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$ **[5]**

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$. **[4]**

(c) Hence write down the exact value of $\sin^2\left(\frac{5\pi}{12}\right)$. **[1]**

Solution

(a) Using de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^6 = \cos 6\theta + i\sin 6\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^6 &= \cos^6\theta + 6\cos^5\theta(i\sin\theta) + 15\cos^4\theta(i\sin\theta)^2 \\ &\quad + 20\cos^3\theta(i\sin\theta)^3 + 15\cos^2\theta(i\sin\theta)^4 \\ &\quad + 6\cos\theta(i\sin\theta)^5 + (i\sin\theta)^6 \\ &= \cos^6\theta + 6i\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta \\ &\quad - 20i\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta \\ &\quad + 6i\cos\theta\sin^5\theta - \sin^6\theta\end{aligned}$$

Equating real parts gives

$$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$$

Now let

$$u = \sin^2\theta \qquad \text{so} \qquad \cos^2\theta = 1 - u$$

Then

$$\begin{aligned}\cos 6\theta &= (1-u)^3 - 15(1-u)^2u + 15(1-u)u^2 - u^3 \\ &= (1 - 3u + 3u^2 - u^3) - 15(u - 2u^2 + u^3) + 15(u^2 - u^3) - u^3 \\ &= 1 - 3u + 3u^2 - u^3 - 15u + 30u^2 - 15u^3 + 15u^2 - 15u^3 - u^3 \\ &= 1 - 18u + 48u^2 - 32u^3\end{aligned}$$

Now factorise:

$$1 - 18u + 48u^2 - 32u^3 = (1 - 2u)(16u^2 - 16u + 1)$$

Substituting back $u = \sin^2\theta$:

$$\cos 6\theta = (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

Hence

$$\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

(b) Let

$$u = \sin^2\theta$$

From part (a), if $\cos 6\theta = 0$, then

$$(1-2u)(16u^2-16u+1)=0$$

So either

$$1-2u=0 \quad \Longrightarrow \quad u=\frac{1}{2}$$

or

$$16u^2-16u+1=0$$

$$\begin{aligned} u &= \frac{16 \pm \sqrt{(-16)^2-4(16)(1)}}{2\cdot 16} \\ &= \frac{16 \pm \sqrt{256-64}}{32} \\ &= \frac{16 \pm \sqrt{192}}{32} \\ &= \frac{16 \pm 8\sqrt{3}}{32} \\ &= \frac{2 \pm \sqrt{3}}{4} \end{aligned}$$

Now

$$\cos\left(6 \times \frac{\pi}{12}\right) = \cos\frac{\pi}{2} = 0$$

so $\sin^2\left(\frac{\pi}{12}\right)$ must be one of

$$\frac{1}{2}, \quad \frac{2+\sqrt{3}}{4}, \quad \frac{2-\sqrt{3}}{4}$$

But

$$0 < \frac{\pi}{12} < \frac{\pi}{4}$$

so

$$\sin^2\left(\frac{\pi}{12}\right) < \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Therefore the correct value is

$$\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

(c) Using part (b):

$$\begin{aligned} \sin^2\left(\frac{5\pi}{12}\right) &= \cos^2\left(\frac{\pi}{12}\right) \\ &= 1-\sin^2\left(\frac{\pi}{12}\right) \\ &= 1-\frac{2-\sqrt{3}}{4} \\ &= \frac{2+\sqrt{3}}{4} \end{aligned}$$

11. **(i)** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$ **[5]**

(ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos\theta}$$

given that $\cos\theta \neq 0$. **[7]**

Solution

(i) Using de Moivre's theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

Expand the left-hand side by the binomial theorem:

$$(\cos\theta + i\sin\theta)^5 = \cos^5\theta + 5\cos^4\theta(i\sin\theta) + 10\cos^3\theta(i\sin\theta)^2 + 10\cos^2\theta(i\sin\theta)^3 + 5\cos\theta(i\sin\theta)^4 + (i\sin\theta)^5$$

Now use

$$i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1, \qquad i^5 = i$$

So

$$(\cos\theta + i\sin\theta)^5 = \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Factor out $\cos\theta$:

$$\cos 5\theta = \cos\theta\left(\cos^4\theta - 10\cos^2\theta\sin^2\theta + 5\sin^4\theta\right)$$

Now use $\cos^2\theta = 1 - \sin^2\theta$:

$$\begin{aligned}\cos 5\theta &= \cos\theta\left((1-\sin^2\theta)^2 - 10(1-\sin^2\theta)\sin^2\theta + 5\sin^4\theta\right)\\ &= \cos\theta\left(1 - 2\sin^2\theta + \sin^4\theta - 10\sin^2\theta + 10\sin^4\theta + 5\sin^4\theta\right)\\ &= \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)\end{aligned}$$

Hence

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$

(ii) From part (i):

$$\frac{\cos 5\theta}{\cos\theta} = 1 - 12\sin^2\theta + 16\sin^4\theta$$

provided $\cos\theta \neq 0$.

Let

$$u = \sin^2\theta$$

Then, since $0 \le \sin^2\theta \le 1$ and $\cos\theta \neq 0$, we have

$$0 \le u < 1$$

So we need the range of

$$f(u) = 1 - 12u + 16u^2 \qquad \text{for } 0 \le u < 1$$

The coefficient of u^2 is positive, so the minimum is at the turning point, where

$$u = \frac{-(-12)}{2 \times 16} = \frac{12}{32} = \frac{3}{8}$$

The minimum value is

$$\begin{aligned} f\left(\frac{3}{8}\right) &= 1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 \\ &= 1 - \frac{36}{8} + 16 \times \frac{9}{64} \\ &= 1 - \frac{9}{2} + \frac{9}{4} \\ &= \frac{4}{4} - \frac{18}{4} + \frac{9}{4} \\ &= -\frac{5}{4} \end{aligned}$$

So the least possible value is

$$-\frac{5}{4}$$

Now consider the largest possible value on $0 \le u < 1$.

As $u \to 1^-$:

$$f(u) \to 1 - 12 + 16 = 5$$

We must check whether 5 is actually attained. Solve

$$\begin{aligned} 1 - 12u + 16u^2 &= 5 \\ 16u^2 - 12u - 4 &= 0 \\ 4u^2 - 3u - 1 &= 0 \\ (4u + 1)(u - 1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{4}$$

But $u = \sin^2\theta$ cannot be negative, and $u = 1$ would mean

$$\sin^2\theta = 1 \quad \Longrightarrow \quad \cos\theta = 0$$

which is excluded.

So 5 is not attained.

Therefore the range of

$$\frac{\cos 5\theta}{\cos \theta}$$

is

$$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos \theta} < 5$$

12. Use de Moivre's theorem to find the constants A, B and C in the identity

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$ **[5]**

Solution

Let

$$z = \cos\theta + i\sin\theta$$

By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$ and $z - z^{-1} = 2i\sin\theta$.

Now write $\sin\theta\cos^4\theta$ in terms of z:

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{z - z^{-1}}{2i}\left(\frac{z + z^{-1}}{2}\right)^4 \\ &= \frac{1}{32i}(z - z^{-1})(z + z^{-1})^4\end{aligned}$$

Expand $(z + z^{-1})^4$:

$$(z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

So

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{32i}(z - z^{-1})(z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}) \\ &= \frac{1}{32i}\big(z^5 + 4z^3 + 6z + 4z^{-1} + z^{-3} \\ &\qquad - z^3 - 4z - 6z^{-1} - 4z^{-3} - z^{-5}\big) \\ &= \frac{1}{32i}\big(z^5 + 3z^3 + 2z - 2z^{-1} - 3z^{-3} - z^{-5}\big) \\ &= \frac{1}{32i}\Big[(z^5 - z^{-5}) + 3(z^3 - z^{-3}) + 2(z - z^{-1})\Big]\end{aligned}$$

Using $z^n - z^{-n} = 2i\sin n\theta$:

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{2}{16}\sin\theta \\ &= \frac{1}{8}\sin\theta + \frac{3}{16}\sin 3\theta + \frac{1}{16}\sin 5\theta\end{aligned}$$

Comparing with

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$

gives

$$A = \frac{1}{8}, \qquad B = \frac{3}{16}, \qquad C = \frac{1}{16}$$

13. **(a)** Using de Moivre's theorem, show that

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

[5]

(b) By using the substitution $\alpha = \left(\frac{\pi}{2} - \theta\right)$, or otherwise, find a similar identity for $\sin^2\theta\cos^4\theta$. **[3]**

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\,\mathrm{d}\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

show that $a = \frac{\pi}{6}$ satisfies this condition and explain why it is the only possible value of a. **[6]**

Solution

(a) Let

$$z = \cos\theta + i\sin\theta$$

By de Moivre's theorem:

$$z^n = \cos n\theta + i\sin n\theta, \qquad z^{-n} = \cos n\theta - i\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

In particular, $z + z^{-1} = 2\cos\theta$ and $z - z^{-1} = 2i\sin\theta$, so

$$\begin{aligned}(z - z^{-1})^4(z + z^{-1})^2 &= (2i\sin\theta)^4(2\cos\theta)^2\\ &= 16i^4\sin^4\theta \times 4\cos^2\theta\\ &= 64\sin^4\theta\cos^2\theta\end{aligned}$$

Now expand the left-hand side, using $(z - z^{-1})(z + z^{-1}) = z^2 - z^{-2}$:

$$\begin{aligned}(z - z^{-1})^4(z + z^{-1})^2 &= (z - z^{-1})^2(z^2 - z^{-2})^2\\ &= (z^2 - 2 + z^{-2})(z^4 - 2 + z^{-4})\\ &= z^6 - 2z^2 + z^{-2} - 2z^4 + 4 - 2z^{-4} + z^2 - 2z^{-2} + z^{-6}\\ &= (z^6 + z^{-6}) - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4\\ &= 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4\end{aligned}$$

Equating the two expressions:

$$64\sin^4\theta\cos^2\theta = 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$$

so

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

Alternative Solution:

Let

$$c = \cos 2\theta$$

Using the double angle formulae:

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-c}{2}, \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2} = \frac{1+c}{2}$$

So

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \left(\frac{1-c}{2}\right)^2\left(\frac{1+c}{2}\right)\\ &= \frac{1}{8}(1-2c+c^2)(1+c)\\ &= \frac{1}{8}(1+c-2c-2c^2+c^2+c^3)\\ &= \frac{1}{8}(1-c-c^2+c^3)\end{aligned}$$

Now use de Moivre's theorem.

From

$$(\cos 2\theta + i\sin 2\theta)^2 = \cos 4\theta + i\sin 4\theta$$

equating real parts gives

$$\cos 4\theta = 2\cos^2 2\theta - 1 = 2c^2 - 1$$

so

$$c^2 = \frac{1+\cos 4\theta}{2}$$

Also, from

$$(\cos 2\theta + i\sin 2\theta)^3 = \cos 6\theta + i\sin 6\theta$$

equating real parts gives

$$\cos 6\theta = 4\cos^3 2\theta - 3\cos 2\theta = 4c^3 - 3c$$

so

$$c^3 = \frac{\cos 6\theta + 3c}{4}$$

Substituting these into $\frac{1}{8}(1-c-c^2+c^3)$:

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \frac{1}{8}\left(1-c-\frac{1+\cos 4\theta}{2}+\frac{\cos 6\theta+3c}{4}\right)\\ &= \frac{1}{8}\left(\frac{4-4c-2-2\cos 4\theta+\cos 6\theta+3c}{4}\right)\\ &= \frac{1}{32}(2-c-2\cos 4\theta+\cos 6\theta)\end{aligned}$$

Since $c = \cos 2\theta$, this is

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$

as required.

(b) Let

$$\alpha = \frac{\pi}{2} - \theta$$

Then

$$\sin\theta = \cos\alpha, \qquad \cos\theta = \sin\alpha$$

so

$$\sin^2\theta\cos^4\theta = \cos^2\alpha\sin^4\alpha = \sin^4\alpha\cos^2\alpha$$

Using part (a):

$$\sin^4\alpha\cos^2\alpha = \frac{1}{32}(\cos 6\alpha - 2\cos 4\alpha - \cos 2\alpha + 2)$$

Now

$$2\alpha = \pi - 2\theta, \quad 4\alpha = 2\pi - 4\theta, \quad 6\alpha = 3\pi - 6\theta$$

so

$$\cos 2\alpha = -\cos 2\theta, \qquad \cos 4\alpha = \cos 4\theta, \qquad \cos 6\alpha = -\cos 6\theta$$

Therefore

$$\sin^2\theta\cos^4\theta = \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$$

So a similar identity is

$$\sin^2\theta\cos^4\theta \equiv \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$$

(c) Let

$$I(a) = \int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\mathrm{d}\theta$$

Using parts (a) and (b):

$$\begin{aligned}\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta &= \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)\\ &\quad + \frac{2}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)\\ &= \frac{1}{32}(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)\end{aligned}$$

Hence

$$\begin{aligned}I(a) &= \frac{1}{32}\int_0^a (-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)\,\mathrm{d}\theta\\ &= \frac{1}{32}\left[-\frac{1}{6}\sin 6\theta - \frac{3}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta + 6\theta\right]_0^a\\ &= \frac{1}{32}\left(6a + \frac{1}{2}\sin 2a - \frac{3}{2}\sin 4a - \frac{1}{6}\sin 6a\right)\end{aligned}$$

We are given that

$$I(a) = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

For $a = \frac{\pi}{6}$:

$$6a = \pi, \qquad \sin 2a = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 4a = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 6a = \sin\pi = 0$$

So

$$\begin{aligned} I\left(\frac{\pi}{6}\right) &= \frac{1}{32}\left(\pi + \frac{1}{2}\times\frac{\sqrt{3}}{2} - \frac{3}{2}\times\frac{\sqrt{3}}{2} - 0\right) \\ &= \frac{1}{32}\left(\pi - \frac{\sqrt{3}}{2}\right) \\ &= \frac{\pi}{32} - \frac{\sqrt{3}}{64} \end{aligned}$$

So $a = \frac{\pi}{6}$ satisfies the equation.

The integrand is $\sin^2\theta\cos^2\theta\left(\sin^2\theta + 2\cos^2\theta\right)$, which is positive for $0 < \theta < \frac{\pi}{2}$, so $I(a)$ increases with a on this interval and the solution is unique.

Therefore

$$a = \frac{\pi}{6}$$